

MITHILESH BIRADAR

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Summary

AI/ML engineer specializing in retrieval-augmented generation, generative AI, and large-scale data systems. Skilled in Python, PyTorch, LangChain, PostgreSQL, AWS, NVIDIA NeMo/NIM, and RAG pipelines. Proven impact across multimodal retrieval, model optimization (52% smaller models, 47% faster inference), and forecasting. Co-presented a RAG system at TD SYNEX Inspire 2025 alongside NVIDIA.

Experience

Clemson University

Oct 2025 - Present

Research Assistant

Clemson, SC

- Optimized the Accelerated Clustered TOPOlogical (ACTOPO) data structure on large-scale meshes (17.4M vertices), resolved complex memory bottlenecks to **reduce cache misses by 99%** and **accelerate computation by 49%**.
- Accepted to **IEEE VIS 2026** conference.

TD SYNEX

May 2025 - Sep 2025

AI Engineer Intern

Greenville, SC

- Engineered a multimodal **GenAI RAG solution** with **NVIDIA**, leveraging LangChain and NVIDIA NeMo to retrieve and reason over structured and unstructured data across multiple databases, accelerating information retrieval for analysts.
- Fine-tuned a query router for multi-source retrieval, achieving **98%+ routing accuracy** with 15 ms routing latency.
- Optimized model inference on an NVIDIA A100 GPU cluster via quantization, reducing model size by 52% and improving **inference latency by 47%** to support near real-time analysis for large-scale multimodal models.
- Built an end-to-end SQL-RAG pipeline automating Excel-to-PostgreSQL ETL workflows, enabling business analysts to uncover data-driven insights and retrieve audit-ready financial data via natural language.

Clemson University

Nov 2024 - Apr 2025

Research Assistant

Clemson, SC

- Researched AI-driven wildfire imagery by analyzing 42 fire/smoke datasets and benchmarking CV/VLM models, including ResNet-50, YOLOv8, CLIP, and Grounding DINO, to advance multimodal fire detection systems.
- Publication:** AI-Driven Fire Imagery Using VLMs and LLMs, Machine Learning with Applications - 2025

Tele Radio

Dec 2023 - Jun 2024

Data Scientist Intern

Pune, MH

- Developed sales forecasting models using Python, Pandas, and Scikit-learn, applying feature engineering and trend analysis on CRM and historical sales data to support demand planning and business decision-making.
- Automated ETL processes using Python and SQL to streamline data integration and reporting, reducing manual effort by 60%, built interactive Power BI dashboards with DAX-based KPIs to provide real-time visibility into sales performance and operational trends.
- Contributed to a CRM modernization project by building Python- and SQL-based workflows to extract, validate, and migrate customer and sales records, ensuring data accuracy and consistency across systems.

TRUMPF Metamation

Jun 2023 - Sep 2023

Machine Learning Engineer Intern

Chennai, TN

- Engineered a real-time computer vision system using PyTorch on Linux to process fused IoT sensor and camera data, reducing inspection cycle time via accelerated defect detection.
- Achieved **99.7% accuracy** in real-time bend-angle detection using convolutional neural networks for quality monitoring.
- Built automated preprocessing and inference pipelines for high-frequency sensor streams, reducing manual inspection, enabling real-time manufacturing quality analysis.

Projects

CosmosPapers - Microsoft - TD SYNEX AI Agents Winner: cosmospapers.com

- Built CosmosPapers, a scalable AI research discovery platform indexing papers from top-tier research conferences, using transformer-based dense embeddings and vector similarity search to power semantic search and personalized recommendations for **110+ active users**.
- Developed the ML analytics and Copilot layer using BERTopic, UMAP, HDBSCAN, and c-TF-IDF to cluster papers into distinct research topics, enabling low-latency trend dashboards and an LLM Copilot for summarization, Q&A, and BibTeX generation.

NASA ZEUS Air Quality Monitoring System

- Built a forecasting pipeline using 5 years of NASA MERRA-2 satellite, weather, and environmental data for air-quality prediction and trend analysis.
- Architected a scalable ML lifecycle on AWS SageMaker to automate model training, versioning, and deployment.

Skills

- Languages:** Python, SQL, C++, C, JavaScript
- AI/ML:** PyTorch, TensorFlow, scikit-learn, LangChain, RAG, NLP, LLMs, VLMs, NVIDIA NeMo/NIM, Search Ranking
- Data/Backend:** PostgreSQL, pgvector, PySpark, REST APIs, ETL/ELT, Pandas, Vector Databases, Power BI
- Cloud/DevOps:** AWS, OCI, Docker, Kubernetes, Linux, Git, CI/CD

Education

Clemson University - M.S. Computer Science (GPA: 3.98)

Aug 2024 - May 2026

MIT Academy of Engineering - B.S. Computer Engineering (GPA: 3.8)

Jun 2020 - May 2024